Answers learning task 12

2026 – 2027

Exercise 7.34: Let X be a geometric random variable with parameter p. Find the maximum likelihood estimator of p based on a random sample of size n.

Answer

$\hat{p} = \frac{1}{\bar{X}}$

Exercise 7.35: Consider the Poisson distribution with parameter λ. Find the maximum likelihood estimator of λ, based on a random sample of size n.

Answer

$\hat{\lambda} = \bar{X}$

Exercise 7.38: Consider the probability density function

$$f(x) = \frac{1}{\theta^2} x e^{-x/\theta}, \quad 0 \leq x < \infty, \quad 0 < \theta < \infty$$

Find the maximum likelihood estimator for θ.

Answer

$\hat{\theta} = \frac{\bar{X}}{2}$

Exercise 7.40: Consider the probability density function

$$f(x) = c(1 + \theta x), \quad -1 \leq x \leq 1$$

(a) Find the value of the constant c.

(b) Show that $\hat{\theta} = 3\bar{X}$ is an unbiased estimator for θ.

Answer

(a) $c = \frac{1}{2}$

(b) $E(\bar{X}) = E(X) = \frac{1}{3}\theta$. So $E(3\bar{X}) = \theta$.

Exercise 7.41: The Rayleigh distribution has probability density function

$$f(x) = \frac{x}{\theta} e^{-x^2/2\theta}, \quad x > 0, \quad 0 < \theta < \infty$$

(a) It can be shown that $E(X^2) = 2\theta$. Use this information to construct an unbiased estimator for θ.

(b) Find the maximum likelihood estimator of θ. Compare your answer to part (a).

Answer

(a) $\hat{\theta} = \frac{\sum_{i=1}^{n} X_i^2}{2n} = \frac{\bar{X^2}}{2}$.

(b) $\hat{\theta} = \frac{\sum_{i=1}^{n} X_i^2}{2n} = \frac{\bar{X^2}}{2}$, the maximum likelihood estimator is identical to the unbiased estimator in part (a).

Exercise 7.42: Let $X_1, X_2, \ldots, X_n$ be uniformly distributed on the interval 0 to a. Recall that the maximum likelihood estimator of a is $\hat{a} = \max(X_i)$.

(a) Argue intuitively why $\hat{a}$ cannot be an unbiased estimator for a.

(b) Suppose that $E(\hat{a}) = na/(n+1)$. Is it reasonable that $\hat{a}$ consistently underestimates a? Show that the bias in the estimator approaches zero as n gets large.

(c) Propose an unbiased estimator for a.

(d) Let $Y = \max(X_i)$. Use the fact that $Y \leq y$ if and only if each $X_i \leq y$ to derive the cumulative distribution function of Y. Then show that the probability density function of Y is

$$f(y) = \begin{cases} \frac{ny^{n-1}}{a^n}, & 0 \leq y \leq a \\ 0, & \text{otherwise} \end{cases}$$

Use this result to show that the maximum likelihood estimator for a is biased.

(e) We have two unbiased estimators for a: the moment estimator $\hat{a}_1 = 2\bar{X}$ and $\hat{a}_2 = [(n+1)/n]\max(X_i)$, where $\max(X_i)$ is the largest observation in a random sample of size n. It can be shown that $V(\hat{a}_1) = a^2/(3n)$ and that $V(\hat{a}_2) = a^2/[n(n+2)]$. Show that if $n > 1$, $\hat{a}_2$ is a better estimator than $\hat{a}_1$. In what sense is it a better estimator of a?

Answer

(a) The largest observation is almost always less than a.

(b) Since $\frac{n}{n+1} < 1$, it consistently underestimates a. The bias is $\frac{na}{n+1} - a = -\frac{a}{n+1} \to 0$ as $n \to \infty$.

(c) $\hat{a} = \frac{n+1}{n} X_{\max}$

(d) $F(y) = \left(\frac{y}{a}\right)^n \Rightarrow f(y) = \frac{d}{dy}F(y) = \frac{ny^{n-1}}{a^n} \Rightarrow E(Y) = \frac{na}{n+1} \Leftrightarrow E\left(\frac{n+1}{n}Y\right) = a$, showing that $Y = \hat{a}_{\text{MLE}}$ is biased.

(e) If $n > 1$, we have that $n(n+2) > 3n \Leftrightarrow V(\hat{a}_1) = a^2/(3n) > a^2 = V(\hat{a}_2)$. Therefore, $\hat{a}_2$ has a smaller variance (i.e., is more precise) than $\hat{a}_1$.